Philipp Benjamin Aretz

Center for Theoretical Physics, Massachusetts Institute of Technology · Cambridge, MA, USA

Research interests

Effective field theory and precision QCD; factorization and resummation for collider observables; medium effects in heavy-ion collisions; mathematical structure of quantum field theory.

Education

PhD in Physics, Massachusetts Institute of Technology, Cambridge, MA, USA — 2024–present

MASt in Mathematics (Part III), University of Cambridge, Cambridge, UK — 2023–2024
Pass with Distinction, 92% — ranked 9th of 294

BSc in Physics, Heidelberg University, Heidelberg, Germany — 2020–2023
1.0 (best possible)
Thesis: *The Keldysh Functional Integral and Bounds on Truncated Expectation Values* (advisor: Prof. Manfred Salmhofer)

Abitur, Goethe-Schule Bochum, Bochum, Germany — 2020
1.0 (best possible)

Research positions

Research Assistant, Heidelberg University, Heidelberg, Germany — 2022–2023
CRC/TRR 191 Symplectic Structures in Geometry, Algebra and Dynamics
Differential delay equations: periodic delay orbits and smooth orbit families. Magnetic billiards: KAM-based extension near convex boundaries.

Publications and preprints

Published

P. B. Aretz, M. Salmhofer, *A rigorous Keldysh functional integral for fermions*, J.Statist.Phys. 193 (2026) 28, arXiv:2508.01787 [math-ph].

P. Albers, **P. Aretz**, I. Seifert, *Families of periodic delay orbits*, J. Differential Equations 410 (2024) 228-250, arXiv:2304.07550 [math].

Preprints

P. B. Aretz, K. Lee, S. T. Schindler, I. W. Stewart, *Factorization of elastic, single, and double diffractive pp scattering*, arXiv:2607.07788 [hep-ph].

Talks

Conference Talks

Diffraction Across Colliders, SCET 2026, Seoul, South Korea — 2026

Posters

Diffraction at the LHC, ESI - New Paradigms for Harnessing Quantum Field Theory at Colliders, Vienna — 2026

Generalised Detectors in Heavy Ion Collisions, ESI - New Paradigms for Harnessing Quantum Field Theory at Colliders, Vienna 2026

Awards and funding

Jennings Prize, University of Cambridge — 2024
DAAD Scholarship, Deutscher Akademischer Austauschdienst — 2023
Wolfson College Scholarship, Wolfson College, Cambridge — 2023

Service and leadership

President MIT Rowing Club, MIT Rowing Club — 2026–present
Treasurer, Edgerton House, MIT — 2025–2026
Organiser, First Year Seminar, Massachusetts Institute of Technology — 2024–2025

Technical skills

Python (NumPy, pandas, TensorFlow); C++ (ROOT, Pythia); Mathematica; MATLAB; LaTeX.

Languages

German (native); English (C2, Cambridge Grade A, TOEFL 117/120); French (basic); Spanish (basic); Latin (Latinum).